

Entry and Exit in a Price Regulated Industry: Gasoline Retailing in Québec*

Jean-François Houde
Department of Economics
Queen's University

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Abstract

The goal of this paper is to empirically examine the effect of price floor regulations on firms' entry and exit decisions. In particular I study the dynamic decisions of gasoline stations using a structural econometric approach. I measure the impact of a specific price floor regulation by observing firms' behavior before and after the implementation of the policy. The data used to estimate the model is a panel of local markets in Québec between 1991 and 2001.

Strategic interaction between local competitors is modeled as a dynamic game between neighboring stations. I estimate the parameters of the model using the conditional choice probability estimator of Hotz and Miller [1993], as suggested by the recent literature on the estimation of dynamic discrete games. I then evaluate the effects of the policy by computing the difference between the continuation values of firms with and without the regulation.

The results show that the price regulation had a significant impact on a firm's option value of staying in the market. A consequence of this is a lower exit probability, which led to a slower re-organization of the industry. Moreover, the impact of the policy is shown to have a larger positive impact on weaker stations (i.e. smaller stations using an older technology).

KEYWORDS: Entry and Exit; Policy Evaluation; Retail Markets.

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Introduction

In many markets, prices are constrained by explicit or implicit floors which are aimed at protecting a group of firms from intense price competition. Agricultural price controls provide insurance to local producers against excessively low prices. Similarly, the Galland Law in France states that no retailer can price a product below its “effective wholesale price”, which excludes discounts from the supplier based on the volume sold. One can also argue that anti-dumping regulations are aimed at protecting local producers by forbidding foreign firms to set price below average variable costs.

The advocates of these regulations typically associate periods of low prices with predatory behavior of a group of firms toward smaller independent retailers. For instance, many States and Canadian Provinces have either implemented or at least examined the possibility of implementing price control regulations or “divorcement” acts in gasoline retail markets, on the basis of alleged predatory pricing practice by major retailers.

However, to the extent that firms are facing significant sunk entry or exit costs, evidence of prices below long run average cost (e.g. during a price war) are not sufficient to conclude that firms are colluding on a predatory strategy. In fact, as discussed extensively in the real option literature (e.g. Dixit and Pindyck [10]), the normal exit threshold (i.e. price under which a firm would exit) is typically well below the long run average cost. In those circumstances, fixing a price floor above this normal exit threshold will have important consequences on firms exit and entry decisions, by raising the option value of being active in the industry.

The goal of the present paper is to examine empirically the effect of price floor regulation on firms’ entry and exit decisions, by measuring the change in firm’s incentive to stay active and enter the industry. In addition, using a structural econometric approach, I infer a measure of firms welfare gain (or loss) associated with the price floor regulation. This research methodology departs significantly from the previous literature on the evaluation of price floor regulations, which had mainly focused on comparing the price setting behavior of firms with and without an price floor¹.

The empirical analysis will be conducted using a panel of gasoline stations in five cities of the province of Québec between 1991 and 2001. Starting in 1997, the provincial government implemented a price floor regulation which fixed a weekly minimum price close to the estimated wholesale price. The regulation also defines a minimum margin of \$0.03 which can be added to the floor if it is proven that a station has set its prices close to the floor level for a long enough period of time. The design and motivations of the regulation are thus very similar to an anti-dumping

¹See for instance Barron and Umbeck [7], Anderson and Johnson [4], Gagné et al. [13]

trade policy.

The model used to evaluate the effect of the policy change is a dynamic entry/exit game between local players in spatially differentiated locations. The model assumes that firms located in geographic clusters interact strategically only with their close neighbors. This restriction reduces the dimension of the problem, and enables the study of entry and exit decisions in medium and large urban markets. By doing so, the proposed methodology departs from the previous literature on entry and exit, which have typically been restricted to small isolated markets (see for instance Bresnahan and Reiss [9], and Mazzeo [20]). Beresteanu and Ellickson [8] study the dynamics of retail markets using a similar methodology, but look at the problem from the chains' perspectives rather than the individual stores.

The estimation of the model uses the conditional choice probability estimator of Hotz and Miller [15], as suggested by the recent literature on the estimation of dynamic discrete games (see Aguirregabiria and Mira [2], Pakes, Ostovsky and Berry [21], Pensendorfer and Schmidh-Dengler [22][23], and Bajari, Benkard and Levin [6]).

The identification of the policy impact is performed by comparing the estimated continuation values of firms before and after the occurrence of the price floor regulation. A similar estimation strategy using an observed policy change in a structural dynamic model has been used in the past among others by Rust and Rothwell [25] to study the impact of a regulation change in the Nuclear Power industry. More recently, Ryan [26] uses an observed environmental regulation change to identify the increase in the operation costs due to more stringent environmental requirements.

The preliminary results indicate that the regulation had a significant impact on firms decisions to enter and exit the market. In particular, firm's are less likely to enter, and more likely to stay active in the industry. These results indicate that incumbent firms are benefiting from this regulation, which acts as a barrier to entry. Furthermore, the regulation had a bigger impact on less efficient stations, compared to the stations who use the more recent technology. This result implies that the price floor successfully protected weaker stations, at the expense of a slower reorganization of the industry.

The rest of the paper is organized as follows. Section 1 discusses the recent trends and organization of the gasoline retail industry. Section 2 introduces the database used in the analysis. Sections 3 and 4 present the model and the estimation strategy. Section 5 presents the empirical analysis. The final section discusses the results and future avenues of research. All Figures and Tables are placed in Appendix.

Table 1: Changes in the number of stations in Québec				
	1989/1995	1995/1998	1998/2001	2001/2003
% Δ stations	-8.45	-10.38	-9.10	-7.41

1 Recent trends in the industry

Over the 1980s and the 1990s, the structure of the North-American gasoline retail industry has evolved toward fewer and larger scale stations. While this trend is common to most retail sectors, it has been particularly pronounced in the gasoline sector in part because of the two oil shocks in the 1970s. The rapid increase of the oil prices has lowered the growth rate of aggregate demand, and increased consumers' sensitivity to price variations. Consequently, the industry started the 1980s with an excess capacity that had to be reduced to allow individual stations to cover their fixed operation costs. Table 1 and Figure 3 illustrate these changes for the province of Québec. In Canada there have been two important waves of exit: 1985 – 1989 and 1993 – 1996. Between 1989 – 1995 in Canada, the number of stations has been reduced by 25%. In Québec the reduction seems to have been delayed. From 1981 to 1991 the number of stations has dropped by 17.5%, while it has been reduced by 30% between 1992 to 2003.

In addition, the structure of the industry has changed over the last 20 years in terms of the types of stations. In particular, the traditional gasoline station, which was offering pump services and automobile repair services, has been slowly replaced by self-service stations offering complementary services (e.g. convenience stores). This tendency reflects the change in the needs of consumers for car repair (from independent garage to automobile dealer garage), as well as the fact that consumers are becoming more sensitive to price variations (since self-service stations typically charge lower prices). Over the 80s, this was beneficial mainly to the independent retailers who were selling unbranded gasoline at lower prices. Consequently, until the mid-90s their market share increased steadily relative to the major brand retailers. Figure 3 shows the evolution of the independents market shares between 1984 and 1996.

Finally, a more recent tendency in the market is the entry of large scale self-service stations. These new stations can store 8 to 10 times the volume of a typical station, through multiple service bays and larger tanks. They typically sell unbranded gasoline, and are often associated with hyper-mart retailers like Wal-Mart or Costco. In the United States the number of independent high volume retailers was estimated to 4411, while the number of hyper-marts offering was closer to 2,434. In 2002, the market share of the hyper-marts was 5.4% of the total consumption of gasoline

Table 2: Average annual sales volume per station in Québec and Ontario

Years	Québec			Ontario	Difference
	Independent <i>ML/Y</i>	Majors <i>ML/Y</i>	Total <i>ML/Y</i>	<i>ML/Y</i>	<i>ML/Y</i>
1992			1, 58		
1995			2, 0	3, 5	1.5
1998	1, 338	2, 837	2, 34	4, 22	1, 88
2002	1, 455	3, 259	2, 64	4, 78	2, 14

Source: CAA and Option Consommateur report (2002)

in the United States. In Canada, Loblaws, Wal-Mart, Costco and Safeway are already present in the market with high volume stations. In Québec however, Costco opened only one of those high volume store, while Loblaw has only recently entered the market.

A consequence of the lack of new entry and slower reorganization of the Québec industry is that the average sales volume of stations is lower in the province than in rest of Canada. Table 2 presents evidence of the gap between the average sales volume of stations from Québec and Ontario. These differences are mainly due to the fact that there are more gasoline stations per capita in Québec, and that the capacity of the stations is lower on average.

1.1 Description of the regulation

The *law on petroleum products* was created in the summer of 1997 and administrated by the Régie de l'énergie du Québec (hereafter the "Board"). This followed the occurrence of an important price war during the summer of 1996, which was associated with predatory pricing behavior by the major retailing chains. The mandate of the Board is threefold:

1. Monitor the gasoline industry, and gather information of prices
2. Determine a weekly floor price or Minimum Estimated Price (MEP)
3. Prevent the occurrence of price wars by imposing a minimum margin regulation in a designated geographic market.

The determination of the MEP is given by the following simple rule which measures the average marginal cost of selling gasoline in each local markets:

$$MEP_{mt} = w_t + tc_{mt} + T_{mt}$$

Where w_t is the minimum wholesale price at the terminal, tc_{mt} is an estimation of the transportation cost to deliver gasoline from the refinery to market m , and T_{mt} is the sum of federal and provincial taxes. The MEP is calculated and posted on the website of the board every Monday.

The role of the MEP is to set a floor price under which a firm can sue its competitor(s) for financial compensation on the basis of excessive and unreasonable commercial practices. This new feature of the law facilitates suing procedures between competitors in the market, in a similar fashion as anti-dumping laws.

In cases where companies repeatedly fail to respect the MEP, the regulation provides the Board the ability to impose an additional minimum margin to the MEP. It allows the board to add \$0.03 to the calculation of the MEP in a specific region after the occurrence of a period of low enough prices. The minimum margin serves two purposes. It first establishes an implicit (or long run) price floor, under which the Board considers that stations are not covering their fixed operating costs. It also enables the Board to compensate indirectly stations after a price war.

The technical working of this policy is as follows. First, after the occurrence of a long enough low price period, a gasoline retailer can ask the Board to investigate evidence of price anomalies. Then, the Board conducts a formal consultation of different groups (retailers, consumer protection groups), in order to evaluate the credibility of the predatory accusation. If the board is convinced of the accusation, it can add \$0.03 to the calculation of the MEP for a certain period of time in a specific geographic zone where the price war occurred. In practice the Board considers that a price is predatory if the margin (price minus the MEP) is below \$0.03 for a long period of time (more than a month).

This minimum margin approximates the average operating cost of a representative station in the province². The geographic zone typically includes all local markets which suffered from the price war. Similarly the length of time for which the minimum margin is applicable is proportional to the length of time of the price war.

This minimum margin regulation has been imposed three times in two markets, St-Jérôme and Quebec city. In St-Jérôme (north of Montréal), it was added to the MEP from 23/04/2002 to 25/02/2003, and again from the 9/12/2003 to the 6/06/2005. The imposition of this price floor followed from the entry of Costco in St-Jérôme in 2000, which drove the market price to the MEP level for more than a year. In Québec city, it was added to the MEP from 3/07/2001 to 3/10/2001. Its imposition followed from a severe price war in the Québec City metropolitan area, during the

²After public consultations, the Board decided that the representative station is a self-service station operating a convenience store, and having an annual sales volume of 350 million liters

fall of 2000. Figures 4 and 5 show the evolution of the minimum surveyed price in both cities before and after the imposition of the minimum margin regulation.

2 Data

The entry and exit decisions of gasoline retailers are examined using a panel of gasoline stations for five cities in the province of Québec: Québec City, Trois-Rivières, Sherbrooke, Chicoutimi, and Drummondville. Québec City, the largest of the group, includes between 200 and 300 stations, while the three others (excluding Drummondville) are middle size cities with between 60 and 100 stations (population between 100,000 and 150,000). Drummondville is the smallest city of the group (population of 40,000 people), and has approximately 40 stations.

The data set is collected by Kent Marketing Consulting, which surveys every urban gasoline market in Canada. This data set includes information on sales volume of active stations in the surveyed markets, as well as firms' characteristics (i.e. location, type of amenities, brand, capacity) for an eleven year period between 1991 and 2001. For all cities except Québec, the information is available for the fourth quarter of each year (a 90 day period). For Québec City, the information is available bimonthly (a 60 day period). For the purpose of the empirical analysis, the sales information will be converted to an average day sales, and only annual information will be used. The final unbalanced panel include 6012 observations, i.e. at most 783 stations for 11 years. More details on the variables used in the empirical analysis are provided in section 5

3 A model of entry and exit into spatially differentiated locations

3.1 Market Environment

The relevant market definition is a metropolitan area or city. Each market m is further divided into a set independent locations of $l_m \in \{1, \dots, L_m\}$, characterized by a major intersection or a street segment³. The definition of a location is aimed at capturing the fact that stations are typically agglomerated along a street or at an intersection. From the customer point of view, stations within the same location are homogeneous in terms of their spatial characteristics. The profitability of a location l is characterized by a discrete state variable z_{lt} , affecting the all stations active in l . Let $Z_t = \{z_{1t}, \dots, z_{L_t}\}$ be the distribution of z 's across locations.

Furthermore, as discussed in section 1, the period studied covers an important reorganization of the industry. This reorganization is associated with a large exit rate from traditional stations, and

³the market m index will be dropped hereafter.

less new entry. In addition, most of the reorganization can be associated with a change in the type of stations active in the market: from conventional stations with service bays, to large capacity stations associated with a convenience store. Since demand for gasoline has been fairly constant over the period, a model with homogeneous firms will not be able to reproduce this non-stationarity. To capture this fact, the environment will include two types of competing technologies, and model the reorganization as a slow change from one dominant technology to the other. More specifically, the two types of firms include a conventional type indexed by c , and gas-bar type index by g . The conventional type refers to a full-service station offering car-repair services as a complementary product. Gas-bar stations, refer to larger capacity stations which don't have service bays, but sell complementary products through a convenience store and/or a car-wash. Let $M_t = \{n_{cjt}, n_{gjt}\}_{j=1}^L$ be the distribution of firms across locations.

Finally the description of the market state is completed by the regulation variable R_t , which is equal to one if the government implemented a price floor regulation. The regulation is perceived to be permanent and unanticipated.

3.2 Individual Firms Problem

The description of the entry and exit decisions closely follows Ericson and Pakes [12], and Pakes, Ostrovsky and Berry [21]. In particular, the game is such that local players decide to enter, stay or exit from each location independently of each other. In other words, as in Bresnahan and Reiss [9] or Mazzeo [20], the model abstracts from any chain or network effects that might affect the decision of a station to exit or enter a specific location. This assumption will not hold if the entry and exit decisions are made solely at the chain level. In the Canadian gasoline market, the reality is likely to lie in between the two extremes since about 20% of stations are owned and operated by national chains. A model in which global players (i.e. major chains) are competing with local players would be more demanding in terms of data and computation, and is left as a future extension.

The entry decision is further restricted by the following assumption:

Assumption 1 *In every period there is a fixed number of potential entrants ($\mathcal{E}$) who can enter each location. Furthermore, every new entrant enters the industry as a type g firm.*

This assumption states that there is a fixed number of potential entrants specific to each location. A more general specification would include an intermediate entry stage where firms decide in which location to become a potential entrant, but would add significant complexities to the model. See Seim [27] for an example of a static location model in which firms simultaneously decide where to locate in a city.

The last part of assumption 1 states that potential entrants can enter only with the “new” technology. This restriction is justified only by the data, since we observe only a small number of entrants entering as type c stations.

The next two assumptions add a private information component to the game, by assuming that potential entrants and incumbents privately observe the realization of the set-up cost and the scrap value of their equipment respectively. These assumptions play the dual role of easing the estimation of the model, and ensuring the existence of pure strategies (bayesian) nash equilibria (see Doraszelski and Satterthwaite [11] for a formal proof, and Seim [27] for a related application of this property).

Assumption 2 *Each potential entrant privately observes the cost of setting up a station, labeled κ . This cost is drawn independently over time and across firms from a cdf F^κ .*

Assumption 3 *An incumbent of type $i \in \{g, c\}$ can choose to exit form the market at the end of the period, and receive a scrap value (net of cleaning costs) equal to ν . Where ν is privately observed, and drawn independently from a cdf F_i^ν .*

Assumption 4 defines more clearly what is considered to be an exit decision in the model: a station is considered to exit if it chooses to physically close, or if it switches technology. This assumption simplifies the model by eliminating the additional option of switching the type of station⁴.

Assumption 4 *Incumbent firms cannot change their type.*

Given the previous set of assumptions, the dynamic decision for an incumbent firm of type $i \in \{c, g\}$ is characterized by a dynamic programming problem in which the state is given by the location index l , the private value shock ν , and the aggregate state $S = \{Z, M, R\}$. The Bellman equation characterizing this decision problem is:

$$v_{il}^I(\nu, S) = \max \left\{ \pi_{il}(S) - F_i + \delta \nu \quad , \quad \pi_{il}(S) - FC_i + \delta E(v_{il}(\nu', S')) \right\}, \quad (1)$$

where $\pi_{il}(S)$ represents the reduced form profit function, FC_i is the fixed operation cost, and δ is the common discount factor. The solution to this problem is characterized by a decision rule, such that the firm will choose to exit ($\chi_{il}(\nu, S) = 0$) if the scrap value ν is larger than the expected continuation value.

⁴In a future extension of the model I will consider a more general assumption to allow type c firms to adopt the new technology.

The decision of a potential entrant is similarly characterized by a dynamic programming problem, for which the state is the privately observed set-up cost (κ) and the state of the market S . That is:

$$v_l^E(\kappa, S) = \max \left\{ 0, -\kappa + \delta E(v_{gl}(\nu', S')) \right\}. \quad (2)$$

The optimal entry rule is such that the firm will choose to enter ($\chi^e(\nu, S) = 1$) if the set-up cost is lower than the expected value of being an incumbent next period (i.e. $\delta E(v_{gl}(\nu', S'))$).

Note that the calculation of the expected continuation values in equations 1 and 2 involves forming an expectation over the equilibrium distribution of active firms M' , and the distribution of local demand states Z' . Since this problem is typically intractable, the next section proposes an approximation method to reduce the dimensionality of the problem.

3.3 Aggregation Assumptions

An important limitation to the model described above is that firms need to take into account the evolution of the market level state variables Z_t and M_t when making their entry/exit decisions. Since these variables are distributions over spatial locations, the problem quickly becomes intractable when L is large. This difficulty is similar to the problem described by Krusell and Smith [18] in a dynamic macro-economic model with heterogeneous agents. In this class of models, agents need to keep track of the aggregate distribution of assets in the economy in order to forecast future level of prices. The solution proposed by Krusell and Smith [18] is to reduce the dimension of the state space by allowing agents to keep track of only the first few moments of the aggregate distribution (typically only the mean). In the same spirit, I approximate the full model by imposing the two following assumptions:

Assumption 5 *Firms behave strategically only with their immediate neighbors active in the same location.*

Assumption 6 *The contribution of the variables Z and M in the reduced form payoff function can be accurately approximated by two single dimension variables z_l and $\tilde{m}_l$.*

Assumption 5 redefines the strategy space to be the set of players' actions in the same location, while assumption 6 reduces the dimension of the aggregate state to two single dimension variables.

The variable $\tilde{m}$ will be defined as the average number of firms in active the other locations, weighted by the distance between those locations. Similarly, the variable z_l will be defined as the weighted average of the neighborhood locations demand shocks. The idea of weighting the neighborhood locations by distances is to approximate the substitution between locations (i.e. the

weights are decreasing in distance). More details on the construction of those variables are provided in the empirical section.

Furthermore, firms are assumed to use the following AR(1) model to form expectation about the future market structure and demand state variable:

$$[z_{lt+1}, \tilde{m}_{lt+1}]' = \mu_0 + \mu_1 R_t + A[z_{lt}, \tilde{m}_{lt}]' + \varepsilon_{lt+1} \quad (3)$$

Where, $A = \begin{pmatrix} a_z & 0 \\ 0 & a_m \end{pmatrix}$

Note that the regulation state enters the forecasting equations, so that the stationary distribution of the variables can be affected by the regulation. Let $\{G_r\}_{r=0,1}$ be the discretized Markov transition matrices describing the transition of the aggregate state variables.

Finally, the payoff relevant aggregate state vector is given by $s = \{n_c, n_g, z, \tilde{m}\}$. The static payoff function for a firm of type i in state s is approximated by:

$$\pi_i(s) = \theta_{iz}z + \theta_{ir}R + \theta_{ii} \log(n_{-i} + 1) + \theta_{i-i} \log(n_{i'} + 1) + \theta_{im}\tilde{m}, \quad (4)$$

where $n_{-i} = n_i - 1$, and $n_{i'}$ is the number of firms of type $i' \neq i$ in the same location. Similar reduced form payoff functions have been used in the empirical literature on entry games following the work of Bresnahan and Reiss [9], as well as more recently by Aguirregabiria and Mira [2] to study the entry and exit decisions of Chilean retailers.

3.4 Equilibrium Definition

Given the previous set of assumptions, the timing of the game is described as followed:

1. At the beginning of each period, potential entrants and incumbents observe the realization of their private shocks κ and ν .
2. The state of the location $\{n_{cjt}, n_{gjt}, z_{jt}, \tilde{m}_{jt}\}$ is revealed, and the regulation state R is announced.
3. Incumbents and potential entrants simultaneously choose their action:
 - If a firm chooses to enter, it pays the set up cost κ , and becomes active in the following period.
 - If an incumbent chooses to exit the market, it stays active in the market until the end of the period, and receives the scrap value of its equipment at the beginning of the following period.

4. Profits are realized, and the period ends.

Following the literature on stationary Markov games, the strategy space is restricted to the set of symmetric Markov strategies. This allows us to remove any calendar time, and condition actions only on payoff relevant state variables (Maskin and Tirole [19]). The main advantage of this assumption, is that it reduces considerably the strategy space, by eliminating any open loop strategies (i.e. strategies that condition on the full history of moves). In particular, this restriction imply that firms condition their strategy on the state of their local market $s = \{n_g, n_c, z, \tilde{m}, R\}$, their privately observed shock κ or ν , as well as their own identity $i \in \{c, g\}$ and their activity state (i.e. incumbent (I) or potential entrant (E)).

Formally, given a set of beliefs $\tilde{p} = \{\tilde{p}_i^a\}_{a \in \{I, E\}, i \in \{g, c\}}$ measuring the subjective probability that the other firms in the market will be active next period, the decisions of an incumbent and a potential entrant are characterized by the following discrete-choice dynamic programming problems:

$$V_i^I(s, \nu | \tilde{p}) = \max_{\chi_i \in \{0, 1\}} \left\{ \pi_i(s) + \delta \nu, \right. \\ \left. \pi_i(s) + \delta \int \sum_{s'} V_i^I(s', \nu' | \tilde{p}), \text{ for } i \in \{g, c\} \Pr(s' | s, \tilde{p}, \chi_i = 1) dF_i^\nu(\nu') \right\} \quad (5)$$

$$V^E(s, \kappa | \tilde{p}) = \max_{\chi^e \in \{0, 1\}} \left\{ 0, -\kappa + \delta \int \sum_{s'} V_g^I(s', \nu' | \tilde{p}) \Pr(s' | s, \tilde{p}, \chi^e = 1) dF_g^\nu(\nu') \right\} \quad (6)$$

Using the symmetry assumption, and the iid assumption on ν and κ , the state transition probability has the following form:

$$\Pr(s' | s, \tilde{p}, \chi_i = 1) = \sum_{(x_g, x_c, e) \in \mathcal{S}_{i, I}} \frac{b(x_i, n_i - 1 | s, \tilde{p}_i^I) b(x_{-i}, n_{-i} | s, \tilde{p}_{-i}^I) b(e, E | s, \tilde{p}^E)}{\times g(z', \tilde{m}' | z, \tilde{m})} \quad (7)$$

$$\Pr(s' | s, \tilde{p}, \chi^e = 1) = \sum_{(x_g, x_c, e) \in \mathcal{S}_{g, E}} \frac{b(x_g, n_g | s, \tilde{p}_g^I) b(x_c, n_c | s, \tilde{p}_c^I) b(e, E - 1 | s, \tilde{p}^E)}{\times g(z', \tilde{m}' | z, \tilde{m})} \quad (8)$$

Where,

$$\mathcal{S}_{i, a} = \left\{ \begin{pmatrix} x_g \\ x_c \\ e \end{pmatrix} \left| \begin{array}{l} n'_g = n_g + e - x_g - I(i = g, a = I), \\ n'_c = n_c - x_c - I(i = c, a = I), \\ (x_g, x_c) \geq 0 \text{ and } 0 \leq e \leq E - I(a = E). \end{array} \right. \right\} \quad (9)$$

$$b(r, n | s, \tilde{p}) = \binom{n}{r} \tilde{p}(s)^r (1 - \tilde{p}(s))^{n-r}$$

Because the state space is discrete, the previous conditional transition probabilities can be written in matrix form. Let Γ_{ia} be this markov transition probability matrix for type i in activity state a . Note that the dimension of this matrix depends on the form of the payoff function because the potential number of firms active in the market is not restricted by the model. However, using the

results of Ericson and Pakes [12], one can show that as long as the profits are bounded, there exists a value $\bar{n}_i$ such that the probability of transiting in states with $n_i > \bar{n}_i$ is zero in equilibrium.

Let $\{\chi_i(s, \nu|\tilde{p}), \chi_i^e(s, \kappa|\tilde{p})\}$ be the optimal threshold decision rules solving the DDP problem in equations 5 and 6. These decision rules are the best-response of each firm, given the beliefs $\tilde{p}$. Alternatively the players' actions can be represented in terms of their best-response choice probabilities. That is, the probability of entering or staying in the market, given beliefs $\tilde{p}$ about the opponents choice probabilities. This representation is obtained by integrating the private value state variables "out" of the optimal decision rules:

$$\Delta_i^I(s|\tilde{p}) = \int \mathcal{I}(\nu < E[v_i^I(s', \nu'|\tilde{p})|s, \chi_i = 1]) dF_i^\nu(\nu) = F_i^\nu(E[v_i^I(s', \nu'|\tilde{p})|s, \chi_i = 1]) \quad (10)$$

$$\Delta^E(s|\tilde{p}) = \int \mathcal{I}(\kappa < \delta E[v_g^I(s', \nu'|\tilde{p})|s, \chi^e = 1]) dF^\kappa(\kappa) = F^\kappa(\delta E[v_g^I(s', \nu'|\tilde{p})|s, \chi^e = 1]), \quad (11)$$

where $\mathcal{I}()$ is an indicator function.

Similarly, integrating the private shocks out of the value functions, the *ex-ante* value functions are given by:

$$v_i^I(s|\tilde{p}) = \int V_i^I(s, \nu, \tilde{\phi}|\tilde{p}) dF_i^\nu(\tilde{\phi}) \quad (12)$$

$$\begin{aligned} &= \pi_i(s) + (1 - \Delta_j^I(s|\tilde{p}))\delta E(\nu|\chi_i = 0) + \Delta_i^I(s|\tilde{p})\delta \sum_{s'} v_j^I(s'|\tilde{p})\Gamma_{iI}(s, s') \\ &= \pi_i(s) + (1 - \Delta_j^I(s|\tilde{p}))\delta E(\nu|\chi_i = 0) + \Delta_i^I(s|\tilde{p})\delta E v_i^I(s|\tilde{p}, \chi_i = 1) \end{aligned}$$

$$v^E(s|\tilde{p}) = \int V^E(s, \tilde{\kappa}|\tilde{p}) dF^\kappa(\tilde{\kappa}) \quad (13)$$

$$\begin{aligned} &= \Delta^E(s|\tilde{p}) \left[-E(\kappa|\chi^e = 1) + \delta \sum_{s'} v_g^I(s'|\tilde{p})\Gamma_E(s, s') \right] \\ &= \Delta^E(s|\tilde{p}) \left[-E(\kappa|\chi^e = 1) + \delta E v_g^I(s|\tilde{p}, \chi^e = 1) \right] \end{aligned}$$

The previous equations provide a representation of the continuation values as a function of the best-response choice probabilities, and the beliefs about the other players' actions. The following defines a Markov Perfect (Bayesian) Nash equilibrium (MPE).

Definition 1 A MPE is a set of probability functions $p = \{p_i^a\}_{i=\{c,g\}, a=\{I,E\}}$ which solves the individual agent problem as defined in equations 5 and 6, and are consistent with the other players action. Alternatively, a MPE is a fixed point of the best-reply probability mapping:

$$\mathbf{p} = \begin{pmatrix} \Delta_g^I(p) \\ \Delta_c^I(p) \\ \Delta^E(p) \end{pmatrix} = \Delta(\mathbf{p}) = \begin{pmatrix} F_g^\nu(E v_g^I(p, \chi_g = 1)) \\ F_c^\nu(E v_c^I(p, \chi_c = 1)) \\ F^\kappa(\delta E v_g^I(p, \chi^e = 1)) \end{pmatrix} \quad (14)$$

As discussed in Aguirregabiria and Mira [2], existence of a MPE is easily proven by Browser’s theorem: a MPE is the solution of a continuous mapping (i.e. the best-response functions) from the space of choice probabilities (compact) to itself. However, one cannot guarantee uniqueness. The econometric methodology section shows how one can recover the parameters of interest while dealing with the multiplicity problem.

4 Estimation Strategy

The model outlined previously predicts a probability of exit and entry, conditional on the state of the market and the type of firms. The data requirement for the estimation is a sample of observed entry and exit rates $Y_{tl} = \{e_{tl}, x_{itl}\}_{j=g,c}$, as well as the observed state variables $X_{tl} = \{n_{gtl}, n_{ctl}, z_{tl}, \tilde{m}_{tl}, R_t\}$. It is also assumed that the econometrician has a consistent estimate of the transition probability matrix $\{G_0, G_1\}$, and the discount factor δ . In principle, the remaining parameters, $\theta = \{\theta_\pi, \theta_F, \gamma\}$, can be estimated either using a short panel including a large number of locations, or a long time series dataset on a single or very few markets⁵.

Recently, a number of authors have suggested using the conditional choice probability (CCP) estimator developed by Hotz and Miller [15] and Hotz et al. [16] to get around the identification problem associated with the multiplicity of equilibria⁶. The starting point of the CCP estimator is the insight that for any dynamic discrete choice model with an additive unobserved utility shock, there exist a one-to-one mapping between the choice probabilities and the value function (i.e. relative to one alternative). Therefore, an unbiased estimate of the value function can be obtained using a consistent first stage estimate of the choice probabilities and the state transition matrix. This permits the computation of the choice probabilities predicted by the model without repeatedly solving the dynamic programming problem. Aguirregabiria and Mira [3] improved the accuracy of this estimator by proposing a K-step algorithm, which sequentially updates the choice probability vector using the previous step parameters estimates.

In a multi-agent framework, this methodology has the advantage of avoiding the multiple equilibria problem. To do so, the following assumption is required:

Assumption 7 *The observed sample is the outcome of a unique pure strategy equilibrium.*

⁵Aguirregabiria and Mira [2] estimate an entry/exit model using a short panel (5 years) of small cities in Chili, while Pensendorfer and Schmidt-Dengler [22] estimate a similar model using quarterly data on two cities in Austria over a 20 years period.

⁶Those papers are: Aguirregabiria and Mira [2], Bajari, Benkard and Levin [6], Pakes, Ostrovsky and Berry [?], and Pensendorfer and Schmidt-Dengler [22][23].

Under this assumption, a consistent conditional choice probability estimator will provide the correct decision rules and equilibrium beliefs used by agents when making their decisions.

To illustrate the details of the estimating algorithm, one also need to assume a specific functional form of the private value shocks. I follow Pakes, Ostrovsky and Berry [?] and assume that the scrap values and the set-up cost are exponentially distributed, with mean $\bar{\nu}_c$ and $\bar{\nu}_g$ respectively for the type c and g firms, and mean entry cost $\bar{\kappa}$.⁷ This functional form assumption has the advantage of providing a closed form expression for the conditional expectation of the scrap value:

$$E(\nu | \nu > Ev_i^I(p)) = Ev_i^I(p) + \bar{\nu}_i$$

Using these assumptions and the linearity of the payoff function (i.e. equation 4), the continuation values can be written in matrix form as a function of the unknown parameters $\theta = \{\theta_\pi, \bar{\nu}_g, \bar{\nu}_c, \bar{\kappa}\}$:

$$\begin{aligned} Ev_i^I(p, \chi_i = 1 | \theta) &= (I - \delta p_i^I \Gamma_{iI})^{-1} \Gamma_{iI} [S_i \theta_{\pi_i} + (1 - p_i^I) \bar{\nu}_i] \\ Ev_g^I(p, \chi^e = 1 | \theta) &= \Gamma_{gE} [S_g \theta_{\pi_g} + (1 - p_g^I) \bar{\nu}_g + p_g^I \delta Ev_g^I(p, \chi_g = 1 | \theta)] \\ \text{Where } S_i &= (R \quad z \quad \log(n_{-i} + 1) \quad \log(n_{i'} + 1) \quad \tilde{m} \quad -\mathbf{1}). \end{aligned}$$

As discussed previously, the estimation of the parameters θ is performed in two steps. First, the conditional choice probabilities are consistently estimated using a reduced-form estimator. Let $\hat{\mathbf{p}} = \{\hat{p}_g^I, \hat{p}_c^I, \hat{p}^e\}$ be the matrix of first-stage estimates. Conditional on these first-step estimates, the model predicts a vector of choice probabilities using the best-response probability mapping:

$$\Delta(\hat{\mathbf{p}} | \theta) = \begin{pmatrix} 1 - \exp\left(-\frac{1}{\bar{\nu}_g} Ev_g^I(\hat{p}, \chi_g = 1 | \theta)\right) \\ 1 - \exp\left(-\frac{1}{\bar{\nu}_c} Ev_c^I(\hat{p}, \chi_c = 1 | \theta)\right) \\ 1 - \exp\left(-\frac{1}{\bar{\kappa}} \delta Ev_g^I(\hat{p}, \chi^e = 1 | \theta)\right) \end{pmatrix} \quad (15)$$

In the second stage, the structural parameters are recovered by minimizing the difference between the predicted choice probabilities, and the observed choice probabilities. This task can be performed by maximizing a pseudo-likelihood function, or minimizing a quadratic loss function (i.e. least-square estimator). As argued by Pesendorfer and Schmidt-Dengler [23], the least-square estimator is potentially more efficient than the PML, because it uses the full set of Nash equilibrium conditions (i.e. equation 14) to estimate the parameters. More specifically, the least-square estimator θ_{LS} solves the following problem:

$$Q(\theta_{LS}) = \min_{\theta} \left(\hat{\mathbf{p}} - \Delta(\hat{\mathbf{p}} | \theta) \right)' \Lambda \left(\hat{\mathbf{p}} - \Delta(\hat{\mathbf{p}} | \theta) \right) \quad (16)$$

⁷The exponential distribution cdf is $F(x) = 1 - \exp\left(-\frac{x}{\sigma}\right)$

Where Λ is a positive definite weighting matrix. Pensendorfer and Schmidt-Dengler [23] derived the optimal asymptotic weighting matrix. In the empirical section, I use an inefficient weighting matrix, and conduct inference using a parametric bootstrapping methodology. More detailed on the estimation is included in the next section.

4.1 Identification of the Policy Effect

As stated in the introduction, the goal of the paper is to measure the impact of a price control regulation on the welfare of incumbents and potential entrants in the market. In the set-up of the model, this involves measuring the following change in the value of being an incumbent and a potential entrant before and after the change in the regulation:

$$D_i^I \equiv v_i^I(R=1)/v_j^I(R=0)$$

$$D^E \equiv v_j^E(R=1)/v^E(R=0)$$

With those estimates, I am able to simulate the evolution of the local markets without the implementation of the regulation, and compare the entry and exit rates under this counterfactual experiment with the observed rates.

5 Empirical Analysis

5.1 Construction of the Local Markets

One major issue associated with the estimation of the theoretical model is the definition of the locations in which firms are competing. The issue is particularly complicated in the case of large spatial markets because each station demand is indirectly affected by the action of all other firms in the same city. On the other hand, modeling the entry and exit decision of firms competing with more than 100 other firms with complicated substitution pattern between them would be infeasible as discussed in section 3.1. To get around this problem, the empirical literature on the gasoline industry has used different definitions of local markets, using some measure of physical distance between firms. For instance, Slade [28] studied the duration of price wars in a single local market of Vancouver, defined as a street segment. Pinske and Slade [24] studied the distribution of ownership contracts between stations of Vancouver assuming various definition of local markets. Their results suggested that the “best” definition of local market (i.e. used in the construction of a spatial weighing matrix) is one in which firms are competing only with their immediate neighbors along a street. More recently, Hastings [14] defined a local market such that “a station competes

with any station within one mile along a surface street or freeway”. One problem with the previous definitions is that the markets are not closed, so that one cannot define meaningful strategies. Therefore, to be in line with the previous literature, and in order to be able to define proper strategies a location must satisfy the three following properties:

1. The distance between stations within a location has to be small enough.
2. Two stations are in the same location if they have at least one street in common.
3. A location must be a close set of firms (i.e. each firm is in only one market), and the locations cannot geographically overlap.

The first two properties ensure that firms within a location share similar spatial characteristics and are homogeneous products, while the last ensures that the game between stations in the same location is defined properly. Iyer and Seetharaman [17] used a similar definition of local markets, constructed by visually clustering stations into local markets.

In order to implement a definition which meets those properties, I use an clustering algorithm which grouped stations according to two criteria: distance and connectivity. In particular, the algorithm iterate on the classification such that the distance between each station and the center of its local market is no greater than 1 mile, and , and stations share at least one street with the others (i.e. the stations are connected). The actual distance criteria can easily be varied if one to create larger or smaller markets. Note that this market definition is time invariant because it uses the total sample of stations who were active in the five cities at some point between 1991 and 2003. In the chosen configuration, the average number of firms per local market is slightly above 3, with a maximum of 15. Figure 6 presents the distribution of firms in those local markets for the five cities.

5.2 Definition of the State Variables

Recall that the state of a location is defined as the state of regulation R_t (i.e. $R_t = 1$ if the price floor regulation is implemented), the number of stations of each type active in the location $\{n_{glt}, n_{clt}\}$, a measure of the market structure in neighboring locations $\tilde{m}_{lt}$, and a measure of the state of the profitability of the location z_{lt} .

To construct these variables, one must choose an appropriate weighing matrix W which weights more heavily closer locations. The choice of the weights is aimed at approximating the elasticity of substitution between two locations. A popular choice in the spatial econometrics literature is to

use a Gaussian weighting function with bandwidth b (e.g. Fotheringham et al. [1]):

$$w_{ij} = \begin{cases} \exp(-1/2(d_{ij}/b)^2) & \text{if } i \neq j \\ 0 & \text{otherwise} \end{cases}$$

Where d_{ij} is the distance between location i and j . The weights are decreasing in d according to a Gaussian curve (i.e. the weights are decreasing at a faster rate when $d_{ij} > b$). Since the markets studied are different in terms of population density, I standardize the weights such that the sum of weights for each location is equal to one. That is:

$$w_{ij} = \frac{\exp(-1/2(d_{ij}/b)^2)}{\sum_{j=1..L} \exp(-1/2(d_{ij}/b)^2)} \quad (17)$$

To construct z_{lt} I use an estimate of the conditional expectation of sales in the local market, conditioning on a set of demographic characteristics. In particular, using the log of sales at the location level (q_{lt}), I estimate the linear relationship:

$$\begin{aligned} q_{lt} &= A_t + B_1 \log(\text{POP}_{lt}) + B_2 \log(\text{TRADE}_l) \\ &\quad \sum_{i=1}^{\# \text{AREA}} C_i \mathcal{I}(l \in \text{AREA}_i) + D_1 \log(n_{glt} + 1) + D_2 \log(n_{clt} + 1) + \varepsilon_{lt} \\ &\quad \text{Where, } \varepsilon_{lt} = \lambda W_t \varepsilon_{lt} + \mu_{lt} \end{aligned} \quad (18)$$

Where $\mathcal{I}(\cdot)$ is an indicator function, and W_t is the weighting matrix defined in equation 17. The categorical variable AREA indexes the location by the type of area (i.e. highway, residential, and/or commercial). The variable POP measures the population in the census tract(s) of location l as measured by the Canadian Census. The variable TRADE measures the average number of retailing establishments (excluding gasoline retailers) in the postal-code group (i.e. FSA) of location l between 1999 and 2001⁸. This last variable was constructed using the Small Area Retail Trade Estimate (or SARTRE) survey, conducted by Statistics Canada. This survey measures the number of retail establishments and the sales for all postal-code groups in Canada on an annual basis. Unfortunately, the data prior to 1999 are not publicly available from Statistics Canada.

Equation 18 was estimated using an iterative GLS procedure (see Anselin [5] for more details). The choice of the bandwidth b in the computation of the weights was set to 2 miles. At this level, the fit of the model (i.e. R^2) is maximized, and the estimated spatial correlation coefficient is lower than 1. Part of the results from estimation 18 are reproduced in table 5.

Using these estimates, the state of the local market is defined as the expectation of sales, conditioning only on the demographic characteristics and the observed profit shocks (i.e. ε_{lt}).

⁸FSA or Forward Sortation Area is defined as the first three components of the postal codes.

$$\begin{aligned}
z_{lt} &= \begin{matrix} 0.934 \\ (0.067) \end{matrix} + \begin{matrix} 0.015R_t \\ (0.0104) \end{matrix} + \begin{matrix} 0.879z_{lt-1} \\ (0.009) \end{matrix} + \begin{matrix} \epsilon_{lt}^z \\ (0.275) \end{matrix} \\
\tilde{m}_{lt} &= \begin{matrix} 0.0275 \\ (0.0086) \end{matrix} + \begin{matrix} 0.016R_t \\ (0.003) \end{matrix} + \begin{matrix} 0.953\tilde{m}_{lt-1} \\ (0.003) \end{matrix} + \begin{matrix} \epsilon_{lt}^{\tilde{m}} \\ (0.0837) \end{matrix}
\end{aligned}$$

Table 3: AR1 results for the aggregate state variables

More specifically, z_{lt} is defined as:

$$z_{lt} = q_{lt} - \hat{D}_1 \log(n_{gl}) - \hat{D}_2 \log(n_{clt}) \quad (19)$$

For markets with no active stations, I do not observed q_{lt} . For those locations the z_{lt} are calculated using their observed characteristics, and the neighborhood locations observed shocks (i.e. weighted by $\hat{\lambda}W$).

Finally, as discussed in section 3.1, the second state variable (i.e. $\tilde{m}_{lt}$) is constructed by calculating the weighted number of active stations in the neighboring locations:

$$\tilde{m}_{lt} = \sum_{l' \neq l} w_{ll'}(n_{cl't} + n_{gl't}) \quad (20)$$

Summary statistics of the variable used in the estimation are included in table 7 below. Table 6 also presents the distribution of entry/exits, and the total number of active firms of each type in the sample over the 11 years.

5.3 First-Stage Estimation

5.3.1 Aggregate States Transition

The initial stage of the estimation routine corresponds to the estimation of the Markov transition matrix for states $(z, \tilde{m})$. As previously discussed in the section 3.1, it is assumed that firms form expectation about those variables using VAR procedure. Equation 3 was estimated by ordinary least-square. The results are reported in table 5.3.1.

Furthermore, these two variables were discretized into 5 grids each (i.e. total of 25 states). The q^{th} grid refers to the q^{th} quantile of the distribution of each variable. Figure 7 presents the kernel smoothed density of the two aggregate state variables for four years. The z_{lt} is slightly increasing over time reflecting the favorable business cycles, while the distribution of the number of stations in the market is clearly decreasing reflecting the reorganization of the industry.

Given this discretization and the estimated parameters, I followed Tauchen [29] method to compute the Markov transition probability matrices (G_0, G_1) (i.e. before and after the implementation of the regulation).

5.3.2 Reduced-Form Choice Probabilities

The choice probabilities were estimated using a simple linear probit model. This approach was chosen, over a non-parametric spell frequency estimator, because the entry/exit decisions are not observed at every discrete states⁹. This is an important limitation in this case because the non-parametric estimator would not yield an estimate of the continuation values at every state. Similar reduced form estimators of the conditional choice probabilities have been used, among others, by Aguirregaberia and Mira [2] and Ryan [26] to estimate dynamic entry/exit games.

More specifically, given a matrix of observed entry and exit choices $Y = \{e_{lt}, x_{glt}, x_{clt}\}$ (i.e. e and x represents the number of entry/exit observations), and a matrix of observed states $X = \{R_t, \log(n_{glt} + 1), \log(n_{clt} + 1), z_{lt}, \tilde{m}_{lt}\}$, the parameters of the reduced form choice probabilities are obtained by solving the following maximum likelihood problem:

$$l(Y|X, \hat{\beta}) = \max_{\theta} \sum_t \sum_l \left\{ e_{lt} \log(\Phi(X_{lt}\theta_e)) + (E - e_{lt}) \log(1 - \Phi(X_{lt}\theta_e)) + \right. \\ \left. \sum_{i=c,g} \left[(n_{ilt} - x_{ilt}) \log(\Phi(X_{lt}\theta_i)) + x_{ilt} \log(1 - \Phi(X_{lt}\theta_i)) \right] \right\}$$

Where $\Phi()$ is the standard normal cdf, and the E the number of potential entrants in each location. This parameter is fixed to 3 in the estimation, which is the maximum number of new entrants observed in the data. The conditional choice probabilities are therefore given by the predicted continuation and entry probabilities obtained using the estimated parameter $\hat{\beta}$. The results of this first-stage choice probability estimation are reproduced in Table 8.

The results from this reduced-form estimation suggest that the regulation had a significant impact on the continuation probability, and the entry probability. In particular, the price regulation reduced the entry probability, and increased significantly the exit probability for two types of firms. Also, the number of active stations of type g reduces significantly the entry probability, but not the number of type c stations. Interestingly, the continuation probability of the type g firms is negatively correlated with the number of active firms of type g , but positively correlated with the number of type c firms. This reinforces the perception that the gas-bar type is seen as a superior technology.

⁹In future revisions of the paper I will consider using a Polynomial approximation of the choice probabilities to obtain more flexible first-stage estimates.

For the type c firms, it appears that the exit probability is much less correlated with the profitability of the locations. This could reflect the fact that these stations might earn a larger fraction of their income from a complementary source (i.e. repair services). Finally, the coefficient associated with z_{jt} has the predicted sign, while the coefficient associated with the market structure variable has the wrong sign and is in-significant in all three equations (i.e. positive instead of negative). This last result suggests that this variable is non-important to predict the continuation/entry choices, so that the aggregate variable z_{lt} would be sufficient to explain firms' profits.

5.4 Second-Stage Estimation

As discussed in section 4, the second stage estimation takes the reduced-form choice probabilities as the equilibrium beliefs about other players' actions, and computes the model predicted choice probabilities. The remaining parameters are then estimated by minimizing the distance between this vector of predicted choice probability and the first-step estimated choice probabilities. Recall that the parameters to be estimated are the set of parameters entering the profits functions for both types ($\theta_\pi = \{\theta_r, \theta_i, \theta_{-i}, \theta_z, \theta_m, FC\}_{i=c,g}$), the average set-up cost ($\bar{\kappa}$), and the average scrap values ($\bar{\nu}_c, \bar{\nu}_g$). However, because I use only the information on the choice to enter and exit, the scale of the profits cannot be identified (i.e. only the relative scale matters). To get around this identification problem, I chose to normalize the average scrap values to 1, and express the continuation values relative to the scrap value of each type. Given these normalizations, the profit parameters are expressed relative to the mean scrap value (i.e. $\theta_\pi = \theta_\pi^0 / \bar{\nu}^0$), and the estimated entry cost is the ratio of the average scrap value of the type g firm and the set-up cost (i.e. $\bar{\kappa} = \bar{\nu}_g^0 / \bar{\kappa}^0$). The 0 superscript identifies the true underlying parameter. Note also that the fixed cost is not separately identified from a constant in the profit function. The value of this parameter should thus be interpreted as the fixed operation cost net of the intercept in the profit function.

Various methods have been proposed to estimate the parameters. I report the results from two of those methods. The first one is the non-linear least-square (NLS) approach proposed by Pesendorfer and Schmidh-Dengler [23], which minimizes the sum of squared difference between the predicted choice probabilities and the observed (or reduced form) choice probabilities. The second approach, advocated by Pakes, Ostrovsky and Berry [21], is a GMM method which minimizes the average differences between the observed choice probabilities and the predicted ones, interacted with a vector of exogenous covariates. More specifically the theoretical and empirical moment

conditions are given by:

$$E(\hat{p}_s - \Delta(\theta, \hat{p})|Z_s) = \frac{1}{\#S} \sum_s Z_s(\hat{p}_s - \Delta_s(\hat{p}, \theta)) = 0. \quad (21)$$

The results reported below use the vector of state variables as instruments. As discussed in Pakes, Ostrovsky and Berry [21], the GMM approach is potentially more reliable in small samples because it is less sensitive to bias from the first-stage choice probabilities.

In both cases, inference is conducted using a parametric bootstrap approach. More specifically, given a set of estimates $\hat{\theta}$, I generate B bootstrap samples of choices $Y_b = \{e, x_c, x_g\}$ by drawing independent private value shocks (ν_c, ν_g, κ) for each active firms in the sample, and using the model to predict the entry and exit decisions at each observed states. This procedure allows us to generate confidence intervals, which take into account the randomness generated by the first-stage choice probability estimates.

Finally, the estimated parameters can be affected by the choice of the state space used to estimate the model. More specifically, although the state space is potentially unbounded (i.e. there is no maximum number of firms active in each location), the model predicts the existence a recurrent subset of the state space, such that the probability of transiting from a state in this subset to a state outside is zero (see Ericson and Pakes [12] for more details). Empirically, one need to assume that the observed sample is drawn from the recurrent set, and make further assumptions about the relevant state space. Pakes, Ostrovsky and Berry [21] suggested to set the state space exactly equal to set of observed states. This assumption has the advantage of reducing the computation cost because only a subset of the possible state space needs to be considered. However, because the first-step estimates (i.e. $\hat{p}$ and $\hat{G}$) predicts that the markets will transit outside the observed states with positive probabilities, the computed transition probabilities might not be consistently estimated if I reduce the state space in this fashion (i.e. the sum of each rows of the transition matrix will not sum to one exactly). I will compare the estimation results from two approaches. The first one fix the state space to the set of observed states, and the second one use the full state space. In particular, the full state space is defined as all possible combinations of the discrete states $s = \{n_g, n_c, z, \tilde{m}, R\}$, where $n_g \in \{0, \dots, \bar{n}_g\}$, $n_c \in \{0, \dots, \bar{n}_c\}$, $z \in \{z_0, \dots, \bar{z}\}$, and $\tilde{m} \in \{\tilde{m}_0, \dots, \tilde{m}\}$. Note that the maximum number of firms of each type is set to the maximum number of observed active firm of each type plus the number of potential entrants (i.e. $E = 3$).

Table 10 presents the results for the restricted state-space using the NLS estimation method. For each, the number of states used in the estimation is 798, and 499 bootstrapping replications have been performed. The results from the full state-space specification are presented in table ??.

For this specification only 19 bootstrap replications have been performed, due to the significant increase in the computation cost (i.e. the size of the state space is 3550).

The parameter estimates are overall precisely estimated. It is especially true for the profit function parameters of the type g firms, and the sunk costs of entry. The payoff function for the type c firms is much less precisely estimated, consistently with the reduced form estimates. The estimated parameters are also quite similar across the estimation methods, and the restrictions on the state space size.

The period profits of the type g firms are negatively affected by the number of competitors of the same type active in the same location. The relation is reverted with respect to the type c firms. This result is not symmetric for the type c firms. For this category of stations, the effect of more competitors of their own type is close to zero, while the effect of the number of competitors of type g is negative. This reveals that the type c firms are weak competitors, either because they have higher cost or lower value for consumers.

The price floor regulation and z_{lt} have both a positive impact on firms' profits, and the regulation have a stronger impact of the type c firms profits. However, as suggested by the reduced form estimation, the market structure variable $\tilde{m}$ does not have a significant impact on the firms profits, and the point estimate has a positive signed. The estimate of the constant is significantly large for the type g firms than for the type c . This can be interpret by the fact that the gas-bar stations have high fixed costs, since they are using a more recent technology and have higher capacity. Alternatively, it can also be the case that selling gasoline is not the main source of revenue for conventional stations. Under both interpretations, this results means that the type c stations are less likely to exit when the state of demand in the local market is low. The gas-bar stations on the other hand react more rapidly to market conditions, since they suffer large losses when the their revenues from gasoline sales are low. Finally, the entry cost is also estimated quite precisely, and the size of the coefficient is large relative to the others.

I next turn to the estimation of the impact of the regulation on firm's value. Table 10 presents the average change in the continuation values for type c and type g firms. The changes are measured in percentage. The mean and the quantiles are over all states in the state space. On average, the regulation has raised the continuation value of firms in the market by about 10% for the type g firms, and by 13% for the type c firms. Since the continuation values are the option value of staying active in the market, this result suggest that incumbents are less likely to exit the market under the regulation state. The fact that the regulation benefited more the conventional stations implies that the policy was effective at protecting smaller less efficient firms in the market. Since those stations

are not using the more recent technology, this means also that the regulation have contributed to the slower reorganization of the Québec industry.

Conclusion

In this paper, I estimated a simple dynamic entry/exit model applied to gasoline stations. Using data before and after the implementation of a price floor regulation, I evaluate the impact of price controls on firms' decisions to exit or enter the market.

A contribution of the paper is to propose a modeling methodology which enables to study dynamic entry/exit games in large urban markets. That is, I reduce the dimensionality of the problem by considering the strategic interaction of players in small homogeneous local markets. To identify the locations, I use a clustering algorithm which group firms into independent locations based on the distance between. Although more work needs to be done to determine the appropriate form of aggregation which defines the state variables and their transition probabilities, I believe this approach could be fruitful to study a wider class of retailing markets.

The preliminary results are summarized as follows. First, the dynamic behavior of firms is found to be very different whether they operate a conventional station with full-service and car-repair shop, or a gas-bar station with convenience store. In particular, gas-bar stations face a higher opportunity cost of staying in the market when the market conditions are not favorable. This reflects higher fixed operation costs, and lower profits from complementary services. Furthermore, the entry cost is very high relative to profits. This implies that the threshold profits over-which stations enter the market must be much higher than the average variable cost, causing firms to delayed their entry decisions.

In terms of the impact of the regulation, I found that it affected positively the profits of both types of stations. This led to higher continuation values, and lower exit probabilities. The impact is also found to be higher on the value of conventional stations. Since those stations are associated with the older inefficient technology, this implies that the regulation protected more effectively the weaker firms in the market. This result is consistent with the fact that the Québec gasoline retail industry has been slower to re-organize than other gasoline markets in Canada and in the United-States.

A Figures and Tables

Type of Convenient store for new and closed stations in Quebec

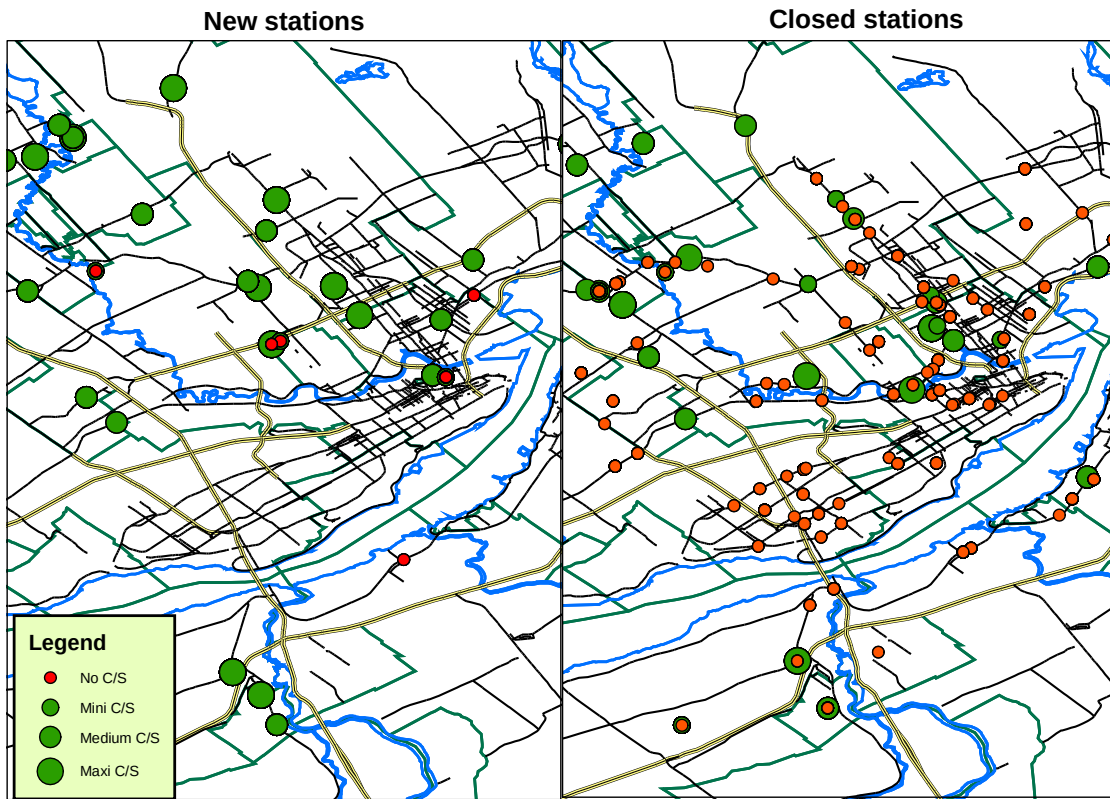


Figure 1: Geographical distribution of closed and newly open gasoline stations in Québec city by type of store

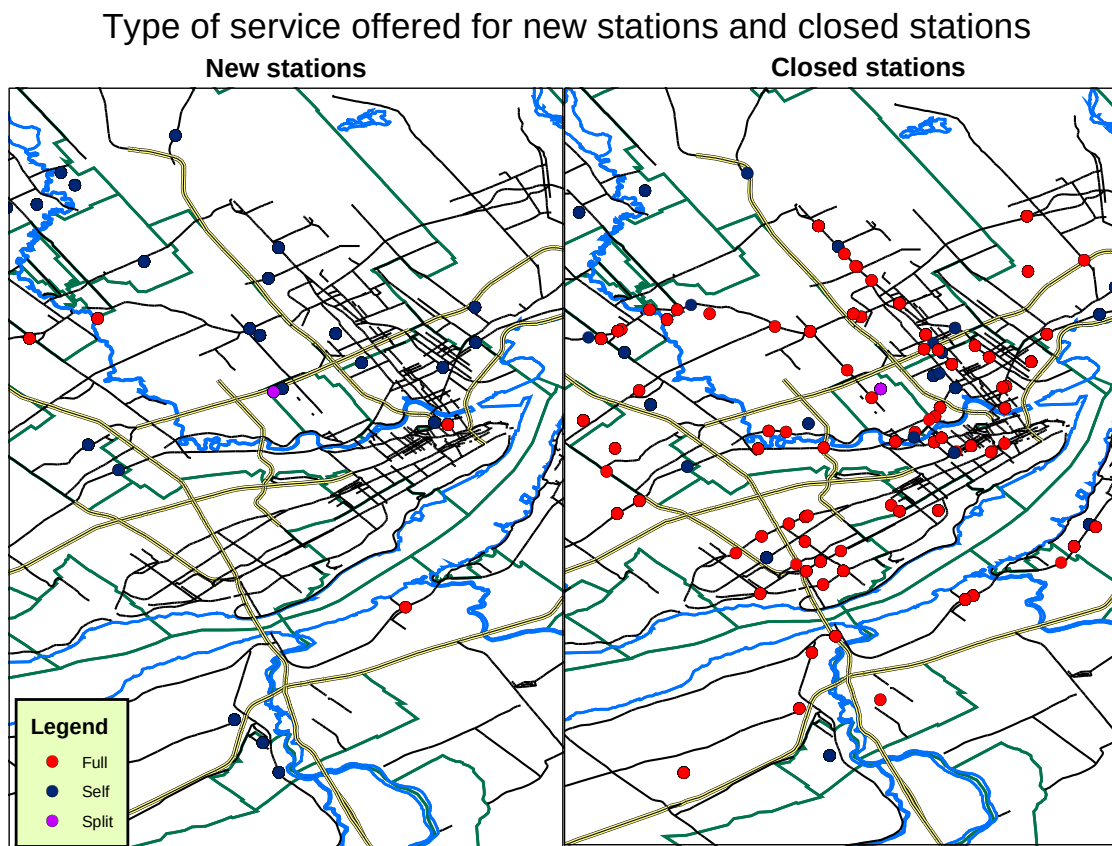


Figure 2: Geographical distribution of closed and newly opened gasoline stations in Québec city by type of service offered

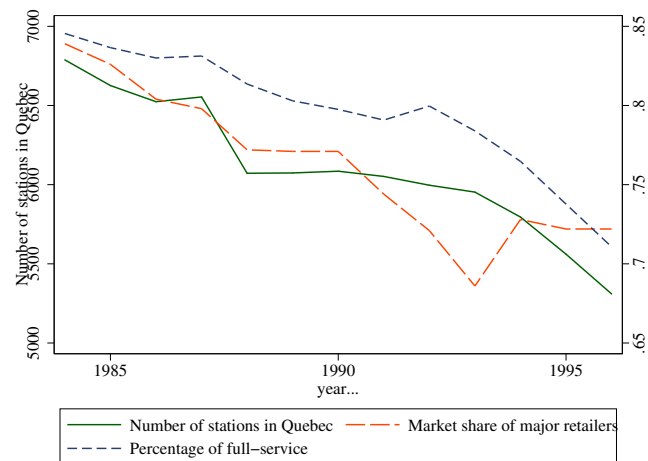


Figure 3: Trends in the province of Québec between 1985 – 1997

Type of Cost	Dollars	Cents per liter
Salary	38000	1.08
Side Benefits	6800	0.19
Uniforms	500	0.01
Capital allowances	17200	0.49
Taxes on petroleum equipments	7000	0.20
Permit	265	0.01
Electricity and heating	9000	0.26
Outdoor maintenance	600	0.02
General maintenance and repair	5100	0.15
Telecommunication	900	0.02
Credit card	8400	0.24
Office furniture	2000	0.06
Inventory losts	3000	0.09
Banking fees	1000	0.03
Insurance	1500	0.04
Professional services	2400	0.07
Publicity	2000	0.06
Total	105665	3,02
Source: Régie de l'Énergie du Québec (1999), D-99-133.		

Table 4: Estimated operation cost

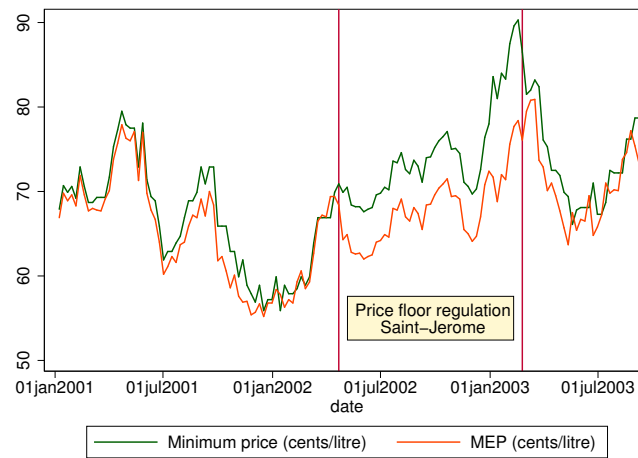


Figure 4: Minimum price in St-Jerome before and after the first price floor regulation

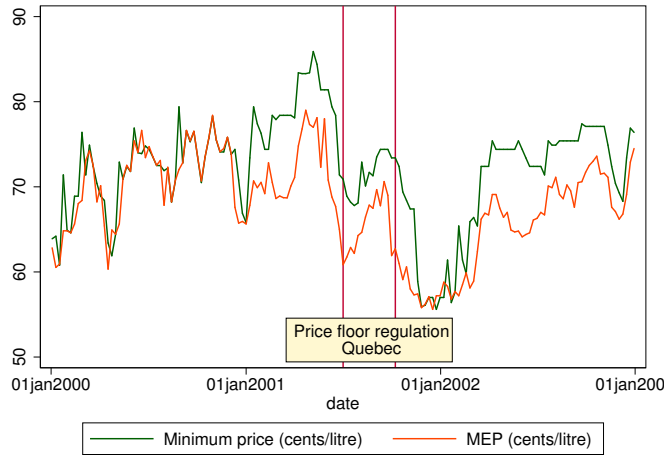


Figure 5: Minimum price in Quebec city before and after the price floor regulation

	Coefficients	Std-Error
TRADE	0.08786	0.01664
POP	0.03791	0.01265
n_g	1.432	0.02717
n_c	0.5879	0.02407
λ	0.7392	
R^2	0.6352	
K	65	
Nb. Obs.	2856	

Table 5: Demand Equation Results (GLS)

Table 6: Number of active firms, entrants, and exits per year

Year	Entrants	Exits (g)	Exits (c)	Incumbents (g)	Incumbents (c)
1991	20	18	24	437	265
1992	15	19	24	439	241
1993	19	13	19	435	217
1994	11	24	15	441	198
1995	7	13	16	428	183
1996	17	23	11	422	167
1997	8	11	10	416	156
1998	8	8	10	413	146
1999	6	15	9	413	136
2000	9	8	9	404	127
2001	6	9	4	405	118

Table 7: Descriptive statistics

	Mean	Std.Dev.	Maximum	Minimum
# Entrants	0.03937	0.2088	3	0
# Exits type g	0.05122	0.2304	2	0
# Exits type c	0.04706	0.2177	2	0
Regulation	0.4545	0.4979	1	0
n_g	1.459	1.211	7	0
n_c	0.6156	0.8955	6	0
Demand state (z)	7.526	0.5881	9.233	3.475
Weighted N ($\tilde{m}$)	2.101	0.462	4.094	0.5394

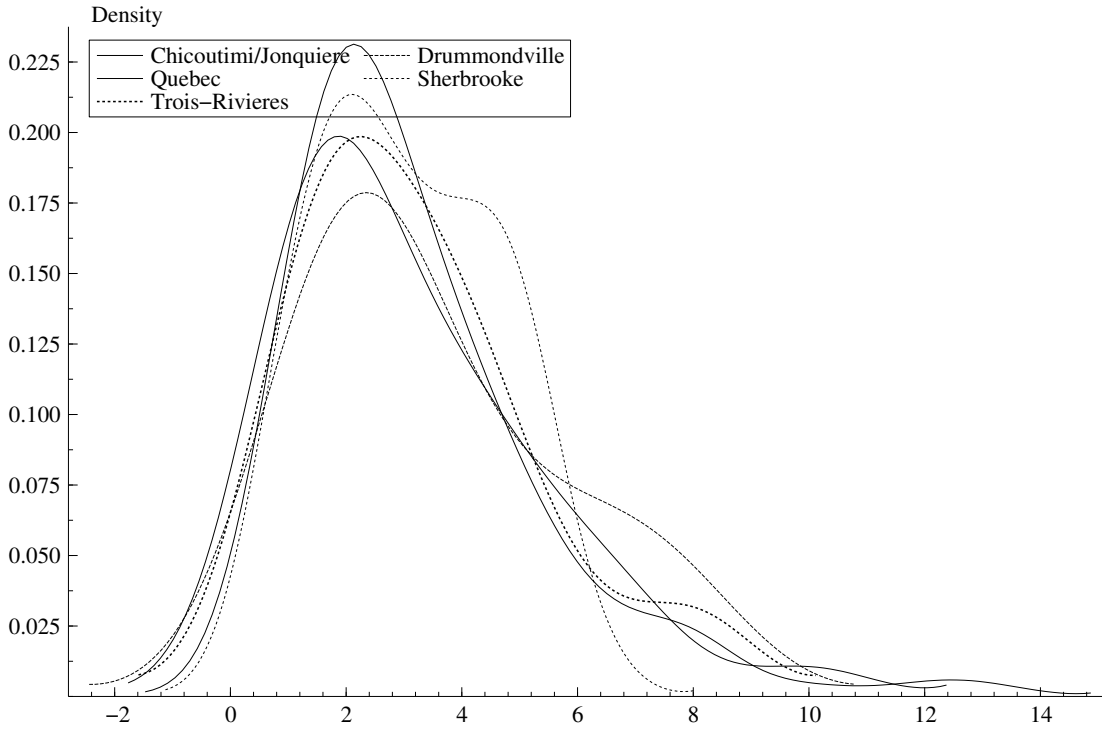


Figure 6: Kernel density of the number of firms by local market

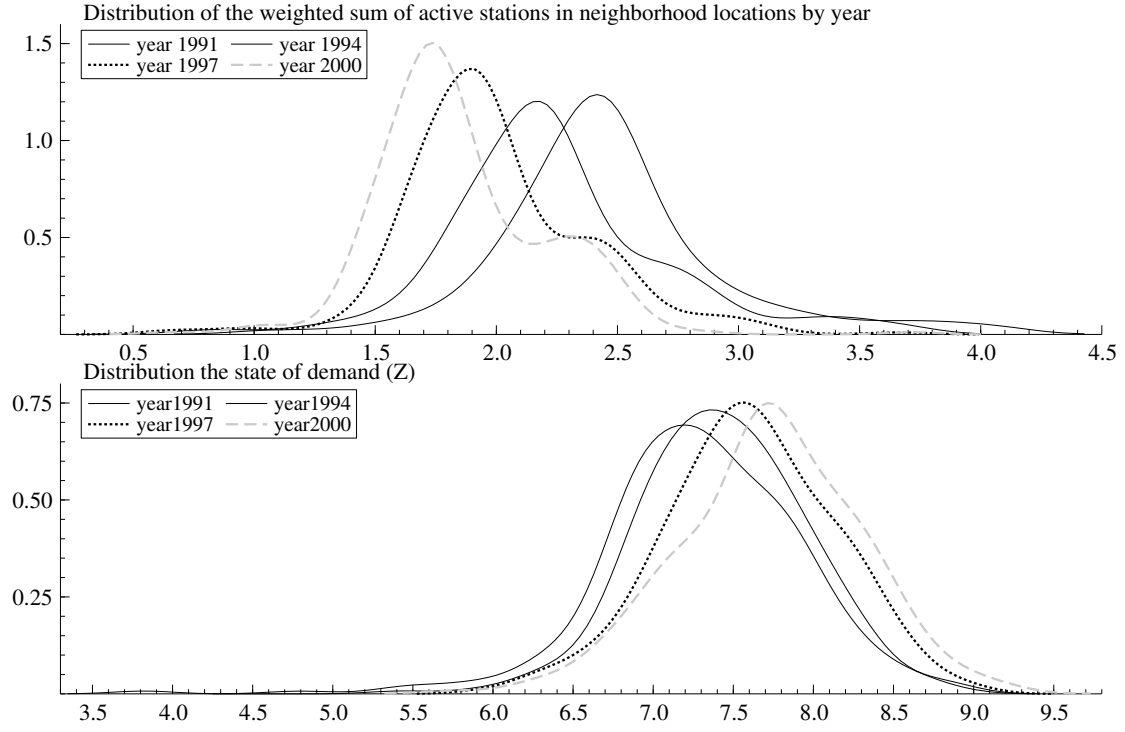


Figure 7: Kernel density of the aggregate state variables

Table 8: Probit results (t-statistics in parenthesis)

	Entry	Stay (g)	Stay (c)
Intercept	-2.763 (-5.1)	-2.15 (-3.8)	0.3086 (0.47)
Regulation	-0.3294 (-4)	0.1703 (2.1)	0.1837 (1.8)
$\log(n_g + 1)$	-0.1476 (-2.1)	-0.2012 (-2)	-0.0491 (-0.59)
$\log(n_c + 1)$	-0.08434 (-1.1)	0.1526 (1.9)	0.03438 (0.26)
z_{lt}	0.1027 (1.6)	0.5076 (7.4)	0.1228 (1.6)
m_{lt}	0.01332 (0.16)	0.1566 (1.9)	0.06794 (0.64)
Nb. Obs.	3124		
LLF	-1816		

Table 9: Estimated Parameters (NLS) on the full state space

	Type g		Type c	
	Estimates	Std-Error	Estimates	Std-Error
θ_R	0.004021	0.01472	0.03442	0.01974
θ_i	-0.03225	0.01585	0.007451	0.01896
θ_{-i}	0.04337	0.01961	-0.01256	0.01961
θ_z	0.4772	0.04675	0.1058	0.04715
θ_m	0.08213	0.03695	0.03499	0.04385
$\bar{\nu}$	1	-	1	-
FC	3.67	0.3663	0.8467	0.3844
$\bar{\kappa}^{1/2}$	20.75	4.383		
Nb. Bootstrap	19			
Policy iterations	1			
Nb. Observations	3124			
Nb. States	1775			

Table 10: Estimated Parameters (NLS) on the restricted state space

	Estimates	Std-Error	5% / 95%
Type g			
θ_R	0.01243	0.01338	[-0.001 ,0.042]
θ_i	-0.02404	0.01708	[-0.045 ,0.011]
θ_{-i}	0.05207	0.02433	[-0.016 ,0.065]
θ_z	0.2914	0.03959	[0.047 ,0.17]
θ_m	0.07586	0.03657	[-0.023 ,0.099]
FC	2.251	0.3292	[0.28 ,1.3]
Type c			
θ_R	0.03576	0.02177	[0.005 ,0.074]
θ_i	0.003003	0.02874	[-0.045 ,0.048]
θ_{-i}	-0.008963	0.02473	[-0.05 ,0.03]
θ_z	0.06354	0.03828	[-0.046 ,0.088]
θ_m	0.03417	0.0425	[-0.059 ,0.083]
FC	0.5252	0.3329	[-0.42 ,0.76]
κ	17.6	1.239	[16 ,20]
Nb. Bootstrap	499		
Policy iterations	1		
Nb. Observations	3124		
Nb. States	399		

Table 11: Estimated Policy Impact (NLS)

	Mean	5% quantile	Median	95% quantile
$\Delta^{\%}V_g$	0.1016	0.06503	0.1021	0.1552
$\Delta^{\%}V_c$	0.1319	0.1165	0.1306	0.1466

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